

Traffic Accident Analysis Dashboard

Data Analytics Mini Project — Project Report

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1. Project Overview and Objective

The **Traffic Accident Analysis Dashboard** is an interactive business intelligence project developed using **Microsoft Excel** and **Microsoft Power BI**. The objective of this project is to analyze traffic accident data across various locations in India and provide meaningful insights into accident cases, injuries, deaths, and accident severity.

The main objective is to demonstrate end-to-end data analytics skills — from data pre-processing and quality assessment in Excel, through data modelling and DAX-based calculations, to an interactive, insight-driven Power BI dashboard — Identify accident-prone regions, compare different accident categories, and understand injury and death statistics through interactive visualizations.

2. Data Sources

Source: Number of Cases Reported and Persons Injured & Died due to Traffic Accidents during 2022

Source Link: <https://ncrb.gov.in/en/ADSI-2022>

Domain: Transportation Analytics (Traffic Safety Analysis)

3. Objectives

- Analyze total traffic accident cases across different locations.
- Compare Road, Railway, and Railway Crossing accidents.
- Identify locations with the highest number of accidents.
- Analyze total injuries and deaths.
- Calculate and visualize the death rate.
- Build an interactive dashboard for decision-making.

4. Attribute (Column / Features) Details

S.No	Attribute (Column Name)	Data Type	Description
1	Sl. No.	Integer	Unique serial number assigned to each record.
2	Location	Text	Name of the State, Union Territory, or City where the accident data is recorded.
3	Location Type	Text	Indicates whether the location is a State, Union Territory, or City.
4	Road Cases	Whole Number	Total number of road accident cases reported.
5	Road Injured	Whole Number	Total number of people injured in road accidents.
6	Road Deaths	Whole Number	Total number of deaths caused by road accidents.
7	Railway Cases	Whole Number	Total number of railway accident cases reported.
8	Railway Injured	Whole Number	Total number of people injured in railway accidents.
9	Railway Deaths	Whole Number	Total number of deaths caused by railway accidents.
10	Railway Crossing Cases	Whole Number	Total number of accidents occurring at railway crossings.
11	Railway Crossing Injured	Whole Number	Total number of people injured in railway crossing accidents.
12	Railway Crossing Deaths	Whole Number	Total number of deaths in railway crossing accidents.
13	Total Traffic Cases	Whole Number	Total number of traffic accident cases, including road, railway, and railway crossing accidents.
14	Total Traffic Injured	Whole Number	Total number of injured persons across all traffic accident categories.
15	Total Traffic Deaths	Whole Number	Total number of deaths across all traffic accident categories.
16	Injury Rate	Decimal / Percentage	Percentage of injured persons relative to the total traffic accident cases.
17	Death Rate	Decimal / Percentage	Percentage of deaths relative to the total traffic accident cases.

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning (Excel)

- Imported the dataset into Microsoft Excel.
- Checked and corrected data types.
- Removed duplicate records.
- Handled missing values where applicable.
- Verified numerical columns.
- Renamed columns for consistency.
- Formatted percentage columns (Injury Rate and Death Rate).
- Converted the dataset into an Excel Table for efficient analysis.

Data Transformation (Power Query)

- Verified data types.
- Renamed columns.
- Removed unnecessary columns (if any).
- Checked for null values.
- Cleaned inconsistent data.
- Applied formatting changes.
- Loaded the cleaned dataset into the Power BI data model.

7. Data Modelling and DAX (Power BI)

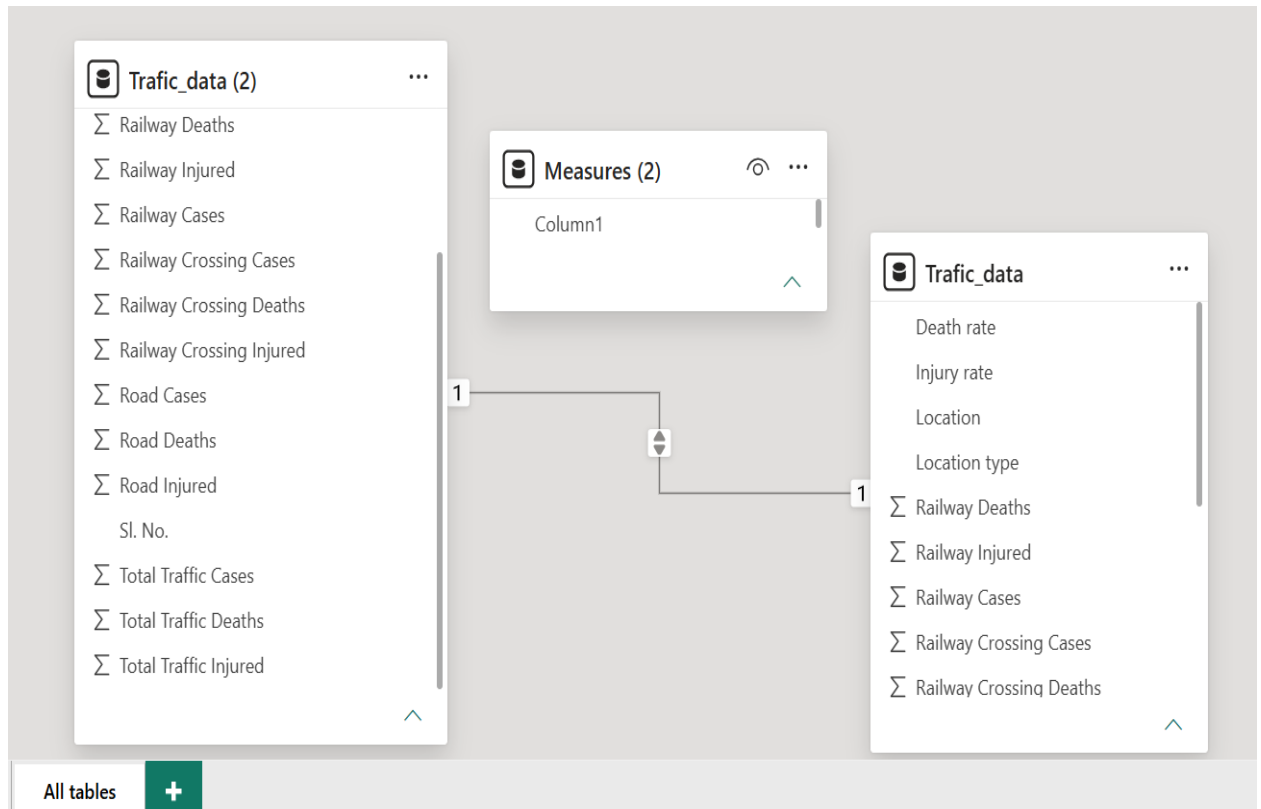
The dataset was imported into Power BI using **Power Query**. As the project consists of a **single fact table (Traffic_Data)**, no relationships between multiple tables were required.

The data model includes accident statistics such as road accidents, railway accidents, railway crossing accidents, injuries, and deaths for various states and cities

Data Model Details

- Imported the cleaned dataset into Power BI.
- Verified and assigned appropriate data types.
- Used a **single-table data model**, so no table relationships were required.

- Ensured all numeric columns were set as **Whole Number** or **Decimal Number**.
- Formatted **Injury Rate** and **Death Rate** as Percentage.
- Optimized the model for reporting and DAX calculations.



Calculated Measures

The following DAX measures were created for dashboard analysis.

Measures	logic	Purpose
Total Accident Cases	Total Cases = SUM('Traffic_Data'[Total Traffic Cases])	Calculates the total number of traffic accident cases.
Total Injured	Total Injured = SUM('Traffic_Data'[Total Traffic Injured])	Calculates the total number of injured persons.
Total Deaths	Total Deaths = SUM('Traffic_Data'[Total Traffic Deaths])	Calculates the total number of deaths

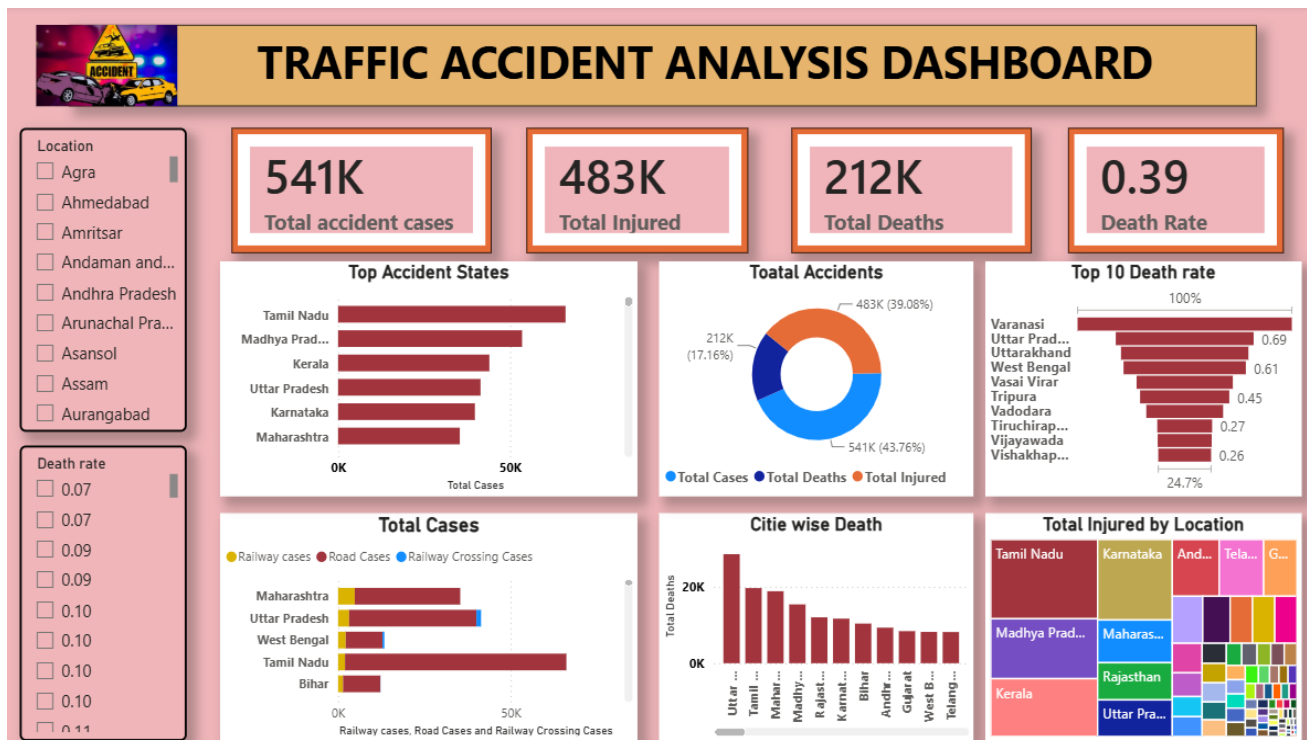
Total Road Cases	Total Road Cases = SUM('Traffic_Data'[Road Cases])	Calculates the total number of road accident cases.
Total Railway Cases	Total Railway Cases = SUM('Traffic_Data'[Railway Cases])	Calculates the total number of railway accident cases.
Total Railway Crossing Cases	Total Railway Crossing Cases = SUM('Traffic_Data'[Railway Crossing Cases])	Calculates the total number of railway crossing accident cases.
Injury Rate	Injury Rate = DIVIDE([Total Injured], [Total Cases])	Calculates the percentage of injured persons relative to total accident cases.
Death Rate	Death Rate = DIVIDE([Total Deaths], [Total Cases])	Calculates the percentage of deaths relative to total accident cases.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

The Power BI dashboard includes several interactive and analytical features that help users understand traffic accident data effectively.

1. KPI Cards
 - Total Accident Cases
 - Total Injured
 - Total Deaths
 - Death Rate
2. Bar Chart – Top Accident States
3. Donut Chart – Total Accident Distribution
4. Funnel Chart – Top 10 Death Rate by Location
5. Line Chart – Accident Type Comparison
6. Column Chart – City-wise Death Analysis
7. Treemap – Total Injured by Location
8. Dynamic Filtering using Slicers



9. Insights & Conclusions

- Tamil Nadu recorded the highest number of accident cases among the displayed states.
- Road accidents contribute significantly more than railway-related accidents.
- Some cities have relatively fewer accident cases but higher death rates, indicating greater accident severity.
- A small number of locations account for a large share of total injuries.
- Interactive filters help compare accident statistics across different locations and death-rate ranges.

9. Conclusion

The **Traffic Accident Analysis Dashboard** successfully transforms raw traffic accident data into meaningful business insights using **Microsoft Excel** and **Power BI**. Through interactive visualizations, KPI cards, DAX measures, and dynamic filtering, the dashboard enables users to monitor accident patterns, compare accident categories, identify high-risk locations, and evaluate injury and death statistics.